

Design and Analysis of a Ladder-Shaped Textile Antenna for SAR-Compliant Driver Fatigue Detection

Siva Keerthi G K

*Department of Electronics and
Communication Engineering Amrita
School Of Engineering,
Amrita Vishwa Vidyapeetham,
Chennai, India.*

Sriram S

*Department of Electronics and
Communication Engineering
Amrita School Of Engineering,
Amrita Vishwa Vidyapeetham,
Chennai, India.*

Rohith Periya Karuppan K

*Department of Electronics and
Communication Engineering Amrita
School Of Engineering,
Amrita Vishwa Vidyapeetham,
Chennai, India.*

Athish Vishnu S R

*Department of Electronics and
Communication Engineering
Amrita School Of Engineering,
Amrita Vishwa Vidyapeetham,
Chennai, India.*

Devisowjanya P

*Department of Electronics and
Communication Engineering
Amrita School Of Engineering,
Amrita Vishwa Vidyapeetham,
Chennai, India.*

Sakthi Abirami B

*Department of Electronics and
Communication Engineering
Sri Sivasubramaniya Nadar College of
Engineering, Chennai, India.*

Abstract— Fatigue of the drivers leading to road accidents is certainly among the top causes, which warrants the creation of unobtrusive monitoring systems for road safety. In this paper, we propose a new textile-based passive RF sensor antenna that is designed and simulated in CST Studio Suite and intends to be integrated into car seats for real-time driver fatigue recognition. The antenna acts by body-coupling effects, where fluctuations in S11 reflection coefficient denoted changes in driver's postures which tend to realize fatigue levels. Printed on a Dacron substrate, the antenna is flexible and durable and can thus be effectively integrated seamlessly into car seats, making it suitable for automotive and wearable applications. The ladder-shaped slot antenna operates with a peak gain of 5.063 dBi at 5 GHz, ensuring very low specific absorption rate (SAR) values that comply with FCC safety guidelines (Total SAR: 0.02757 W/kg, Max 10g SAR: 1.38851 W/kg). S11 shifts are monitored by the system, which can then process them and trigger an alert. This work demonstrates the great potential of passive RF sensing for fatigue detection and provides a non-intrusive, real-time road safety solution for drivers.

Keywords—Textile antenna, body-coupling effect, passive RF sensor, SAR compliance; body Area Network.

I. INTRODUCTION

Driver drowsiness is one of the main causes of road accidents globally, responsible for a large number of fatal crashes. Traditional drowsiness detection systems, including camera-based eye-tracking systems and physiological sensors, are prone to drawbacks such as high computational needs, privacy issues, and user discomfort. In an effort to counter these problems, this paper suggests a non-intrusive, textile-based RF sensing solution to track driver drowsiness.

The system to be implemented uses a ladder-shaped slot antenna [1], integrated into the car seat, which acts as a passive RF sensor [2]–[6]. The basic operating mechanism is on the basis of body-coupling effects, where in the antenna couples with the driver's torso, head. When the driver is completely

alert and sitting upright, there are small air gaps [7][8], visualised in Fig.1(a), between the body and the antenna, resulting in stable S11 reflection coefficient values. But with increasing fatigue, the driver slowly drifts away from the seat, creating air gaps, visualised in Fig.1(a), (b) and resulting in a shift in the resonance frequency and power absorption characteristics. Greater body movements reflect serious fatigue, accentuating these differences further. These dynamic S11 changes are measured and processed to establish the driver's fatigue status.

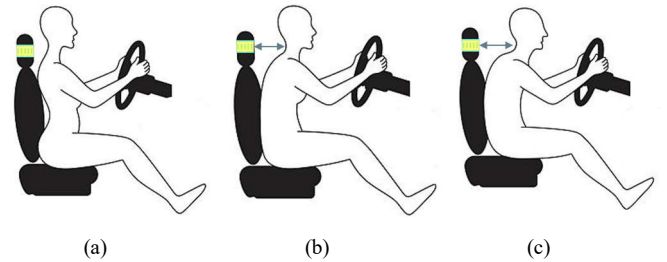


Fig. 1. (a) No air-gaps (Driver is active) (b) Small air-gaps (Driver is mildly fatigue) (c) Large air-gaps (Driver is critically Fatigue)

One of the salient features of the antenna design is its substrate material, Dacron, with a relative permittivity of 3. Dacron is generally employed in wearable and automotive applications because of its mechanical flexibility, light weight, and textile integration compatibility. Its low permittivity further reduces signal distortion, facilitating reliable passive RF sensing and making the antenna to be readily embedded along the lining of car seat [9].

The antenna works at an operating frequency of 5 GHz, exhibiting a peak gain of 5.063 dBi. Ascertaining safety standard compliance, the Specific Absorption Rate (SAR) is thoroughly checked to ensure FCC guidelines compliance with a total SAR of 0.02757 W/kg and a peak 10g SAR of 1.38851 W/kg [10],[11]. The SAR analysis is done by modelling a Human Tissue Phantom (worn a cloth for realistic

simulation) which is kept in contact with the antenna (patch-in-contact with the Human Tissue model). The input power of 0.025 W is fixed for effective operation while not surpassing regulatory standards.

To analyse the response of the antenna experimentally, Vector Network Analysis (VNA) could be used to monitor real-time changes in S11 parameters. If a critical shift in threshold is noticed, the system may issue an alert using an embedded microcontroller-based module, which may also start a buzzer or alert system. The details of how this integration is done are explained in section-V.

This paper is organized as follows: Section-II is a Literature Survey of similar works on driver fatigue detection and RF sensing methods. Section-III discusses the Antenna Design, explaining the structural optimizations for improved performance. Section-IV is the Simulation Results, which verify the antenna's gain, bandwidth, and SAR compliance. Section-V discusses the System Setup, including integrating RF sensing within the car seat and with on-board electronics (if necessary). Lastly, Section-VI concludes with main findings and future directions towards enhancing the system.

This study proves that RF-based textile antennas present a practical, non-invasive solution for detecting fatigue in real-time, capable of enhancing automotive safety standards and mitigating fatigue-caused road accidents.

II. LITERATURE SURVEY

The article proposes design of a ladder-shaped microstrip patch antenna for mobile TV in the UHF band. Their antenna has optimal impedance matching and miniaturization through the use of slots and coaxial feeding. Although they are dealing with TV reception, their slot-based tuning mechanism is in line with my antenna design. Their bandwidth and efficiency enhancement results can be used to optimize my textile-based driver fatigue detection antenna installed in car seats [1].

The article presents passive resonant sensors, such as electromagnetic, acoustic, and magnetoelastic sensors, for environmental monitoring and wearable sensing applications. They are seen to possess benefits, including energy autonomy and remote sensing. This is reflected in our paper, where a textile antenna is used as a passive RF sensor for driver fatigue monitoring. The idea of resonance-based sensing in this article justifies the utilization of reflection coefficient variations in monitoring drowsiness in drivers, hence justifying the passive monitoring in my project [3].

This paper presents a passive UHF RFID-based wearable knitted compression sensor for on-body respiratory tracking. The crucial aspect for our problem is that it focuses on minimizing human body-induced losses and on-body radiation efficiency, following our textile antenna requirement for proper operation near human tissue. Their use of conductive knitted fabric for flexibility and RF sheet resistance extraction to optimize material properties are aspects which could further minimize our antenna substrate selection and integration technique. Their SAR evaluation (0.25 W/kg) also validates our safety compliance concern,

further underlining the need to maintain SAR levels within regulative limits within wearable devices [4].

This paper reviews chip-less RFID sensors with cost-effectiveness, durability, and maintenance-free designs. It discusses the classification of sensors, detection mechanisms, and biomedical applications. This study gives strength to the applicability of passive RF sensing for fatigue detection, which we have in our paper. The information regarding the interactions of electromagnetic radiation with human tissue and what material one should use will help me improve the performance and accuracy of my Dacron-based ladder-shaped antenna [5].

This paper proposes a wireless passive SCG sensor employing RF backscattering at 2.4 GHz and 4.8 GHz for cardiac monitoring. Featuring a dual-band antenna, a PVDF sensor, and an LCP substrate in order to miniaturize the system and enhance biocompatibility capabilities, it is composed of several components. While these systems could be quite effective with respect to vibrometry, my project incorporates these body-coupling effects for fatigue detection. Moreover, their RF backscattering method appeared to be a source of insight into low-power data transmission this could help in improving processing signals basically in my design [6].

The intermodulation sensor is battery-less and chip-less. The passive microstrip patch antenna is used for wireless crack detection. The proposed solution offers improved range and low complexity in clutter-resistant sensing. Mitigating drivers' fatigue can cost-effective reliability and resilience in the project. Intermodulation signal extraction of the above could enhance the processor signal detecting subtle body movement variations [7].

This paper investigates ways to reduce human body interference in UHF RFID antennas through the different approaches related to impedance stability and selection of appropriate material. This very much aligns with our work that is implementing Dacron substrate and slot-based tuning to provide similar mitigation for the effect on the fatigue-monitoring activities. The major highlights of the study concern impedance matching and substrate optimization as two priority factors of body-worn antenna design. In these lines, fine-tuning the matching network and adding protective layers will contribute to the improved performance and efficiency of the antenna [8].

The study focuses on the minimization of SAR in mobile antennas through the use of ground plane grooves; the work reduces surface currents and electric field absorption in human tissues. Changing the ground structure enables SAR to be cut down by a better than 30% margin, which was balanced against retained efficiency. Beneficial to our textile-based fatigue detection antenna, it provides insights into the interaction of antenna with ground structure in respect of the influence of SAR factor and efficiency, for design

optimization of safety without compromise in performance [10].

This study deals with a flexible dual-band wearable antenna for WBAN scenarios. It deals with bending, body loading, and SAR compliance-friendly antenna characteristics, which reflect our own concerns—efficient performance—within the car seat environment. Though the functionality of the antenna has been optimized at both 2.45 GHz and 5.2 GHz frequencies, the methodology regarding antenna SAR validation and flexible substrate selection suits the overall aim of ensuring the system's safe and effective performance in proximity to a human body. Significantly contributing to efficient antenna performance is the light on techniques that have been outlined in the study, which can further be inculcated to make our fatigue monitoring system user-friendly [11].

III. ANTENNA DESIGN

The antenna that has been designed possesses a ladder-type patch in order to have a maximum surface area with good radiation characteristics. There are five slots cut in the rectangular patch in order to change the shape of the antenna to achieve a better impedance matching and larger bandwidth. The slots incorporated change the current distribution effectively, and the performance of the antenna in gain and resonance at the operating frequency improves [1].

The antenna is designed with an operating frequency of 5 GHz. From reading various research papers, the substrate material was selected to be Dacron because it can be applied in wearable and automotive technology. Dacron has a relative permittivity of 3 and a density of 1380 kg/m³, making it easy to incorporate into car seat material [8]. The patch and ground plane are designed using annealed copper for efficient conductivity. The dimensions of the antenna are shown in Table I, and its front and rear views are illustrated in Fig. 2(a), (b), respectively.

TABLE I. ANTENNA DIMENSIONS

Parameters	Dimensions (mm)
Copper Thickness	0.035
Substrate Height	1.5
Ground Length (lg)	40
Ground Width (wg)	56
Patch Length (lp)	32.86
Patch Width (wp)	55
Feed Width (wf)	3.3
Slot Length (sl)	16.43(1/2 the length of patch)
Slot Width (sw)	2

The ladder-type slots play a very significant role in governing the performance of the antenna. The incorporation of the slots brings in new resonant modes, providing impedance bandwidth but maintaining compactness. The slots assist in reducing the surface current density so radiation efficiency is maximized.

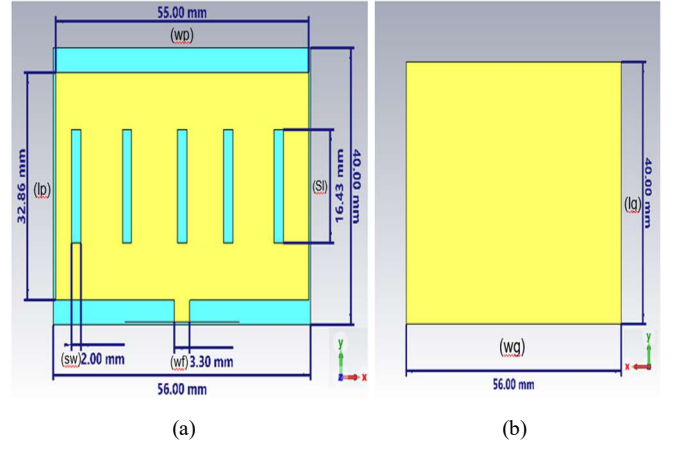


Fig. 2. Ladder-shaped Antenna (a) Front view (b) Rear View

IV. SIMULATION RESULTS

The ladder-shaped antenna performance was measured under two scenarios:

A. Free-Space Operation (antenna with no human intervention)

In the free space, the antenna was simulated without external interference of any dielectric material. The results indicate that the antenna successfully works at 5 GHz, with the corresponding parameters presented in Table II.

TABLE II. FREE SPACE ANTENNA PARAMETERS

Parameters	Values
Gain	5.063 dBi
VSWR	1.229
Reflection Coefficient (S11)	-19.748 dB
Total Efficiency	-0.8948 dB (81.6%)
Radiation Efficiency	-0.7757 dB (83.2%)

Reflection coefficient determines minimum impedance mismatch and thus power transmission efficiency. The antenna shows excellent total efficiency and radiation efficiency and has strong directional radiation patterns.

Besides the structural changes, the reflection coefficient (S11 parameter), voltage standing wave ratio (VSWR), total efficiency, and radiation efficiency of the antenna were examined to assess its performance. The reflection coefficient (S11) at the operating frequency is -19.748 dB, which is good impedance matching, as illustrated in Fig 3. The VSWR, which is a measure of the quality of impedance matching, is 1.229, as illustrated in Fig 4.

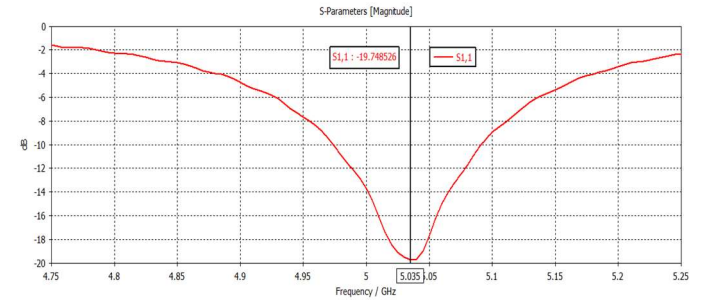


Fig. 3. S11 parameter

The far-field radiation characteristics were evaluated in 3D, 2D, and 1D views, as shown in Fig 5, 6, and 7, respectively.

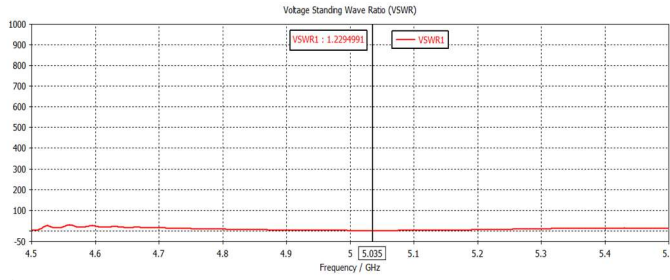


Fig. 4. VSWR

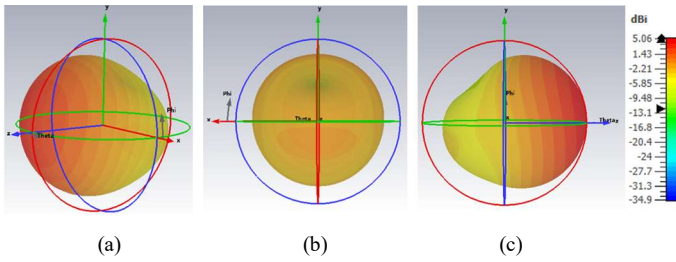


Fig. 5. 3D far-field radiation (a) XYZ plane (b) XY plane (c) YZ plane

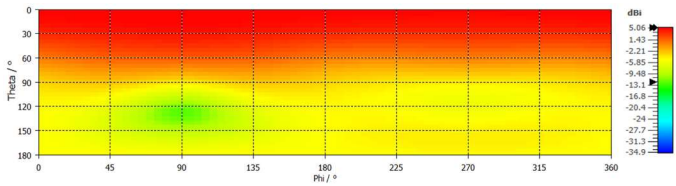


Fig. 6. 2D far-field radiation

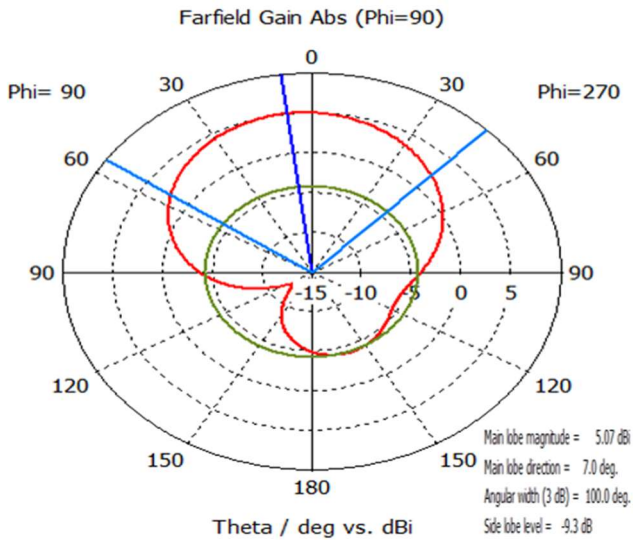


Fig. 7. 1D far-field radiation

B. On-Body Operation (antenna on the body of the driver, simulated above a human phantom model)

To analyze the performance in practice, a human phantom model was established, as shown in Fig. 8. (a), which includes independent layers representing skin, fat, and muscle, placed one above the other as in Fig. 8. (b). These layers have unique values of permittivity, conductivity, and density [10]. The front and rear views of the model are

illustrated in Fig. 9, respectively. After integrating the driver's seat, the antenna experiences body-loading effects and, therefore, undergoes changes in its operating parameters, as given in Table 2.

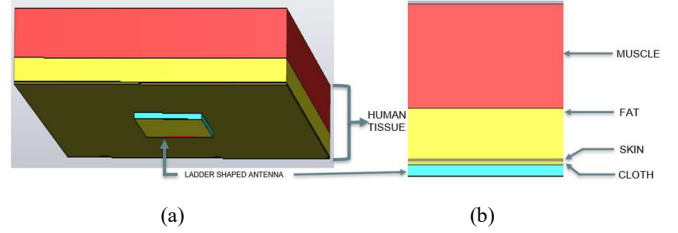


Fig. 8. (a) Human Phantom(tissue) Model(placed directly on the patch) (b) Cross-section of human tissue on Antenna

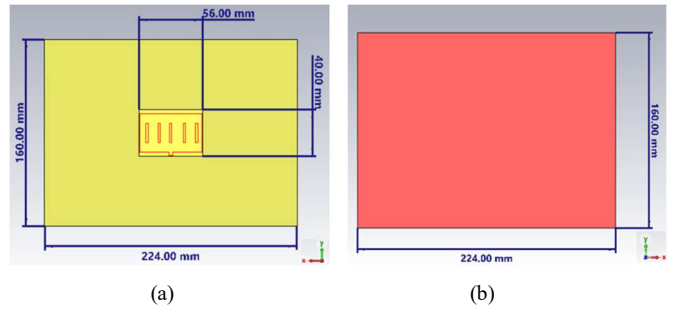


Fig. 9. XY-Planar Views of Human Tissue(a) Front view (b) Rear View

The human tissue causes return loss to increase, together with a shift in the resonance frequency, which subsequently influences impedance matching. But the antenna continues to possess high gain and bandwidth, thereby ensuring fatigue monitoring is performed reliably.

SAR Calculation Results For compliance with safety, SAR was calculated according to IEEE/IEC 62704-1 standards. By reducing the input power to 0.025 W, the antenna keeps SAR in acceptable limits and is represented visually in Fig.10. The major SAR conclusions are given in Table III:

TABLE III. SAR RATINGS

Parameters	Values
Total SAR	0.0275707 W/kg
Maximum Point SAR	20.1956 W/kg
Max SAR (10g)	1.38851 W/kg
Accepted Power	0.0224577 W
Absorbed Power	0.0181313 W

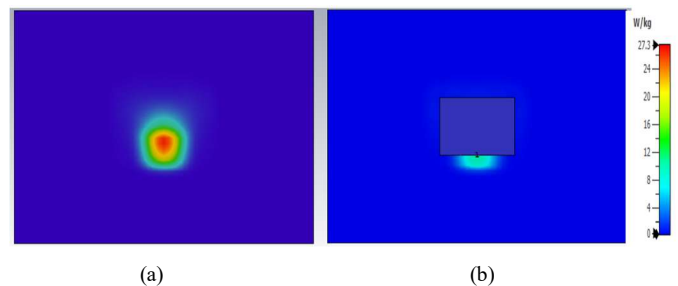


Fig. 10. SAR topology (a) Front view (b) Rear view

The results verify that the antenna is safe when it is used near the human body, with little RF exposure risk but efficient signal transmission. In addition, the improved ladder-like structure is very important in guaranteeing operating efficiency despite the loading effect of human tissue.

V. SYSTEM SETUP

The antenna is positioned to be incorporated into the driver's car seat, more precisely in the headrest or backrest area, thereby enabling effective monitoring through its proximity to the driver's body. The head rest area acts as an ideal locus for the antenna to be placed since it is the head that makes readily-movements in-accordance with whether the driver is drowsy. For the purpose of a seamless integration of the antenna, the car seat has to be upholstered with Dacron material, which is the primary textile medium in this configuration.

In real-world measurement and analysis, a Vector Network Analyzer (VNA) can be employed to measure variations in the reflection coefficient. The antenna can further be integrated with an embedded system through a UFL/SMA connector. An embedded system can be utilized to code for automatic detection of perceptible variations in the reflection coefficient, which may be indicative of driver drowsiness. In the event such variation is detected, the system can trigger an alarm system, such as exciting buzzers and strobe lights, to alert the driver.

This strategy guarantees effective data gathering simultaneously with the added safety feature through the addition of real-time monitoring supported by an automatic alert system.

V.CONCLUSION

The simulation results confirm the performance of the antenna with uniform gain, reflection coefficient, and SAR values within tolerable limits. The ability of the antenna to sense the change in reflection coefficient when near the driver indicates its suitability for integration with in-vehicle warning systems. With a UFL/SMA interface, the antenna can transmit real-time information to an external system, which can initiate safety functions like alarm systems in the event of drowsiness.

Future research studies will focus on optimizing the design of the antenna to improve its sensitivity in fatigue

detection, improve its interaction with human tissues, and incorporate machine learning algorithms to realize a high degree of precision in predicting fatigue. This work demonstrates a significant step towards improving road safety using advanced antenna technology and embedded systems.

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